



Belgium wins lawsuit vs. Facebook

The country of Belgium has won a lawsuit against Facebook, claiming the company does not respect privacy. <http://digit.in/belgvFB>



Russian troll factory

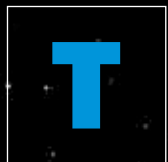
A Russian troll factory apparently had a \$125 million monthly budget to mess with the 2016 US election. <http://digit.in/trollfact>

CLEARING UP SPACE JUNK

With an increasing number of satellites being abandoned, and more being launched to replace them, the amount of junk in space is increasing.

Aditya Madanapalle

aditya@digit.in



he space around Earth is littered with upper stages of launch vehicles, as well as derelict

satellites. There are over 500,000 pieces of space debris, moving at speeds in excess of 28,000 kilometers per hour!

At the end of its operational period, the energy required to increase the altitude of a satellite is less than the energy required to deorbit the satellite. Because of this, defunct satellites are moved into what is known as the junk orbit or the graveyard orbit, to reduce the

possibility of collisions and creating more space debris.

A Chinese anti-satellite missile test in 2007 was responsible for creating the largest amount of space debris in a single event. Over 3,000 pieces of debris were tracked, with each piece being golf ball sized or larger. The debris created by the test passed six kilometers away from the international space station in 2011. In 2009, a commercial satellite built in the US collided with a Russian military satellite, creating over 2,000 pieces of trackable debris pieces.

With each collision, the chances of further collision increases. A chain reaction of hypervelocity collisions, as seen in the movie *Gravity*, is known as the Kessler effect. It is critical for future spaceflight operations that

efforts are taken to reduce orbital debris. The deorbit approaches include active and passive propulsion systems on board the spacecraft, external removal by other satellites, and ground based solutions as well.

NETS AND HARPOONS

This might sound like a fishing expedition, and is an active debris removal approach. Tiny CubeSats deploy a net through a harpoon, to capture the debris. The satellite subsequently deorbits along with the catch. The Surrey Satellite Technology Ltd and Airbus have developed two Debris-Sats to test out this approach.

DEORBITING SAIL

There are two types of sails, active sails and passive sails. Solar sails use the energy from the sun to steer the satellite towards the ground. A drag-sail, is a passive propulsion system that uses the particles in the upper reaches of the atmosphere, to make the satellite fall back to the Earth faster. Both sails can be incorporated into the satellites themselves. The technologies for deploying the sails are susceptible to jamming, and there is an altitude limit to the dragsail.

LASER BLASTERS

One of the ground based plans by NASA to hasten the reentry of derelict satellites, is to reduce their velocity with ground based lasers. The laser broom would slow down the debris by 1 mm per second, and would require firing over multiple passes. Recently, researchers from the Chinese Air Force Engineering University conducted a study, and found that it was viable to use high energy lasers from a space station to destroy orbital debris, and break it down into smaller, less harmful pieces.

PROPULSION SYSTEMS

An active propulsion system can be incorporated into the satellite, specifically for actively deorbiting the satellite. This approach is more viable with the smaller nanosatellites,